MINERALS

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Minerals

- Minerals are essential for the normal growth and maintenance of the body.
    - **MACROELEMENTS:** If the daily requirement is ***more than 100 mg***, they are called major elements or macrominerals or macroelements.
    - **TRACE ELEMENTS:** If the requirement of certain minerals is ***less than 100 mg/day***, they are known as microelements or microminerals or trace elements.

Classification

Major elements	Trace elements
1. Calcium	1. Iron
2. Magnesium	2. Iodine
3. Phosphorus	3. Copper
4. Sodium	4. Manganese
5. Potassium	5. Zinc
6. Chloride	6. Molybdenum
7. Sulfur.	7. Selenium
	8. Fluoride.

IRON

IRON

- Iron is one of the most important trace elements in the body.

- The World Health Organization (WHO) has estimated that the global prevalence of anemia to be ~**24.8%**. The most common causes of anemia include nutritional deficiencies, particularly **iron deficiency**.

- Total iron content in a human varies approximately from 3 gm to 4 gm.

Distribution of Iron in the body

Protein/Enzyme	*Iron content (in mg)*	*% of total*
Haemoproteins		
• Haemoglobin	2500	60–70
• Myoglobin	400	5–10
• Haem enzymes:		
Catalase and peroxidase	2–3	1
Organo-iron compounds		
• Cytochromes	4–5	1
Storage iron		
• Ferritin	300–700	10–15
• Haemosiderin		
(Non-haem protein)		
Transferrin		
(Non-haem protein)	6–8	1


CALCIUM
Ca
Ca
Ca
Ca
Ca

CALCIUM (Ca)

- Calcium is an important mineral mainly found in bone and teeth.

Body distribution:

- Calcium forms about 1% to 1.5% of an average adult's body weight.
- 99% of which is seen in bones and teeth as carbonate or phosphate of calcium.
- 1% of total body Ca occurs in extracellular fluid and soft tissues.

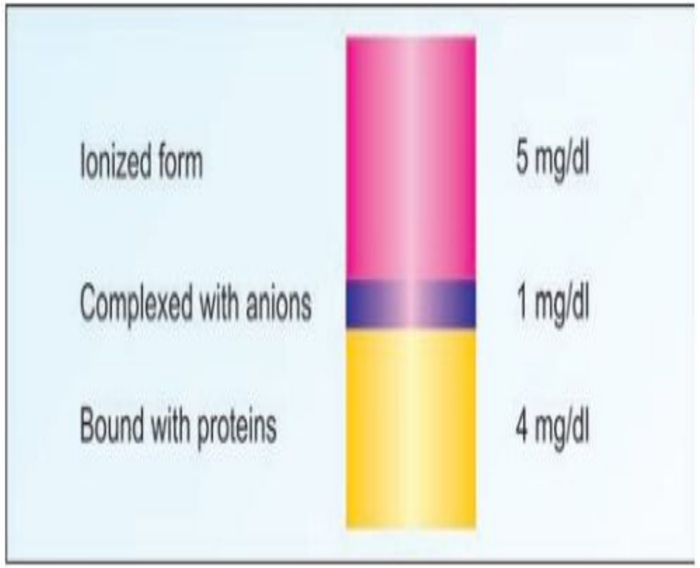
Different forms of calcium in PLASMA

- The normal level of plasma calcium is 9-11 mg/dl.
- The calcium in plasma is of 2 forms:

  - Non diffusible or Plasma protein bound calcium. Albumin is the major protein with which calcium is bound.

  - Diffusible:

1. Ionised calcium(Ca^{2+}): 40% of total calcium is in ionised form and is physiologically active form.
2. Complexed calcium, it is probably complexed with plasma anions such as citrate and phosphate.



Dietary sources of Calcium:

- Milk and Yoghurt is a good source for calcium
- Cheese
- Egg-yolk
- Beans
- Lentils
- Nuts. Of all nuts, almonds are among the highest in calcium. Figs
- Fish and vegetables are medium sources for calcium
- Cereals (wheat, rice) contain only small amount of calcium.

FUNCTIONS

- Calcification of Bones and Teeth.
- Calcium plays a role in blood coagulation: Calcium is known as factor IV in blood coagulation cascade.
- Nerve conduction: Calcium is necessary for transmission of nerve impulses from presynaptic to postsynaptic region .
- Calcium mediates excitation and contraction of muscle fibers.
- Myocardium: Ca^{++} prolongs systole. In hypercalcemia, cardiac arrest is seen in systole. This fact should be kept in mind when calcium is administered intravenously. It should be given very slowly.
- Secretion of hormones: Calcium mediates secretion of insulin, parathyroid hormone, calcitonin, vasopressin, etc. from the cells.
- Activation of enzymes.

PHOSPHORUS

- Phosphate plays a crucial role in cell structure, signaling, and metabolism.
- Phosphate is found both as mineral phosphate(inorganic phosphate) and organic phosphate (phosphoric esters).

- **Food Sources:** Foods rich in phosphorus content are cheese, milk, nuts, organ meats, egg.

- **Body Distribution:**
  - Total body phosphate is about 500-700 g.
  - 85 % is found in bones and teeth
  - 14% in soft tissues
  - 1% is found in ECF, less than 1% circulating in serum
  - Phosphate is mainly an intracellular ion and is seen in all cells.

- **Daily requirement:** About 1 g of phosphate is required to be taken in the diet daily.

Absorption:

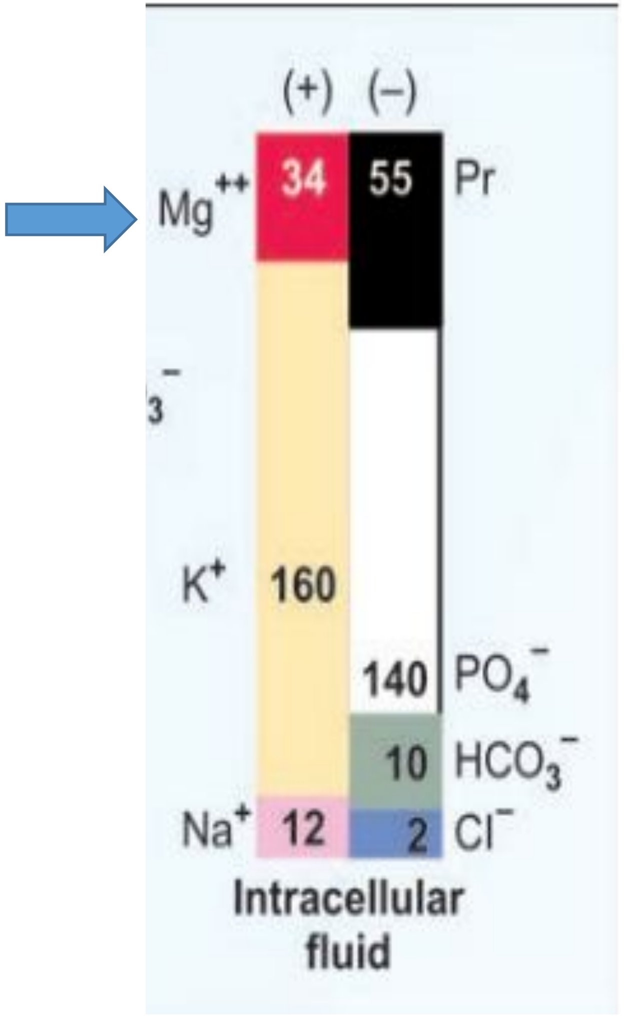
- 90 per cent of daily dietary phosphate is absorbed.
- The absorption is stimulated by both PTH and Vit. D3.
- Phosphorus holds an inverse relationship with calcium. An excess of serum calcium or phosphate results in the excretion of the other by the kidney.
- PTH increases calcium and phosphate release from the bone and decreases loss of calcium and increases loss of phosphate in the urine.

Functions of Phosphate Ions

1. Formation of bone and teeth.
2. Production of high energy phosphate compounds such as ATP, CTP, GTP, creatine phosphate, etc.
3. Synthesis of nucleoside co-enzymes such as NAD and NADP.
4. DNA and RNA synthesis, where phosphodiester linkages form the backbone of the structure.
5. Formation of phosphate esters such as glucose-6-phosphate, phospholipids.
6. Formation of phosphoproteins, e.g. casein.
7. Activation of enzymes by phosphorylation.
8. Phosphate buffer system in blood.

MAGNESIUM (Mg++)

- Magnesium is the fourth most abundant cation in the body and second most prevalent intracellular cation.
- Distribution: Approximately 2/3 occurs in bones, 1 per cent in EC fluid and remainder in soft tissues.
- Requirement:
  - The requirement is about 400 mg/day for men and 300 mg/day for women. Doses above 600 mg may cause diarrhea.
  - More is required during lactation.
  - Major sources are cereals, beans, leafy vegetables and fish.



Functions of Magnesium

1. Mg^{++} is the (cofactor) activator of many enzymes requiring ATP.
   - Alkaline phosphatase, hexokinase, fructokinase, phosphofructokinase, adenyl cyclase, cAMP dependent kinases, etc. need magnesium.

2. Neuromuscular irritability: Mg exerts an effect on neuromuscular irritability similar to that of Ca^{++}, high levels depress nerve conduction and low levels may produce tetany (hypomagnesaemic tetany).

3. As Constituent of Bones and Teeth: About 70 per cent of body magnesium is present as in bones, dental enamel and dentin.

SULFUR

- Sulphur is an essential element.
- The sulphur is made available to the body by the proteins containing methionine, cystine or cysteine. These amino acids contain sulphur.
- Also available from S-containing B vitamins, Thiamine (TPP), coenzyme A, Lipoic acid and biotin.
- Certain sulpholipids and glycoproteins (mucoitin and chondroitin sulphuric acid) also provide Sulphur.
- Sulphur as free element cannot be utilized.

Functions of Sulfur

1. Sulfur containing amino acids are important constituents of body proteins. The disulfide bridges keep polypeptide units together, e.g. insulin, immunoglobulins.

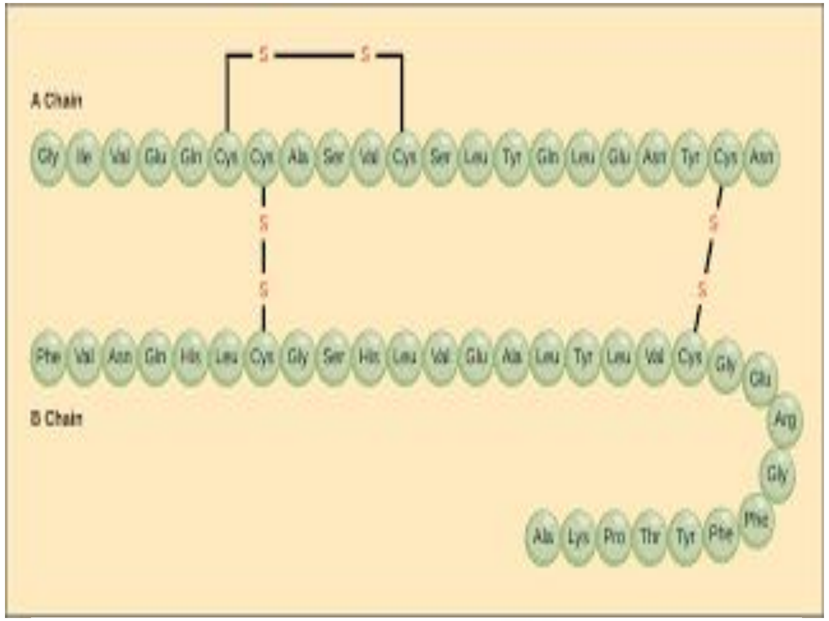
2. Chondroitin sulfates are seen in cartilage and bone.

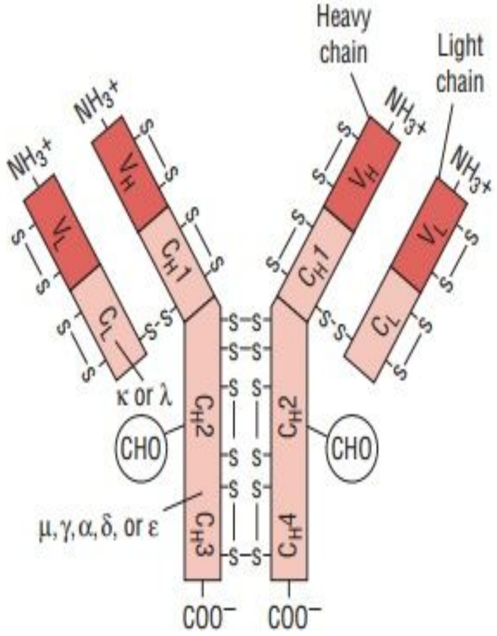
3. Keratin is rich in sulfur, and is present in hair and nails.

4. Many enzymes and peptides contain -SH group at the active site, e.g. glutathione.

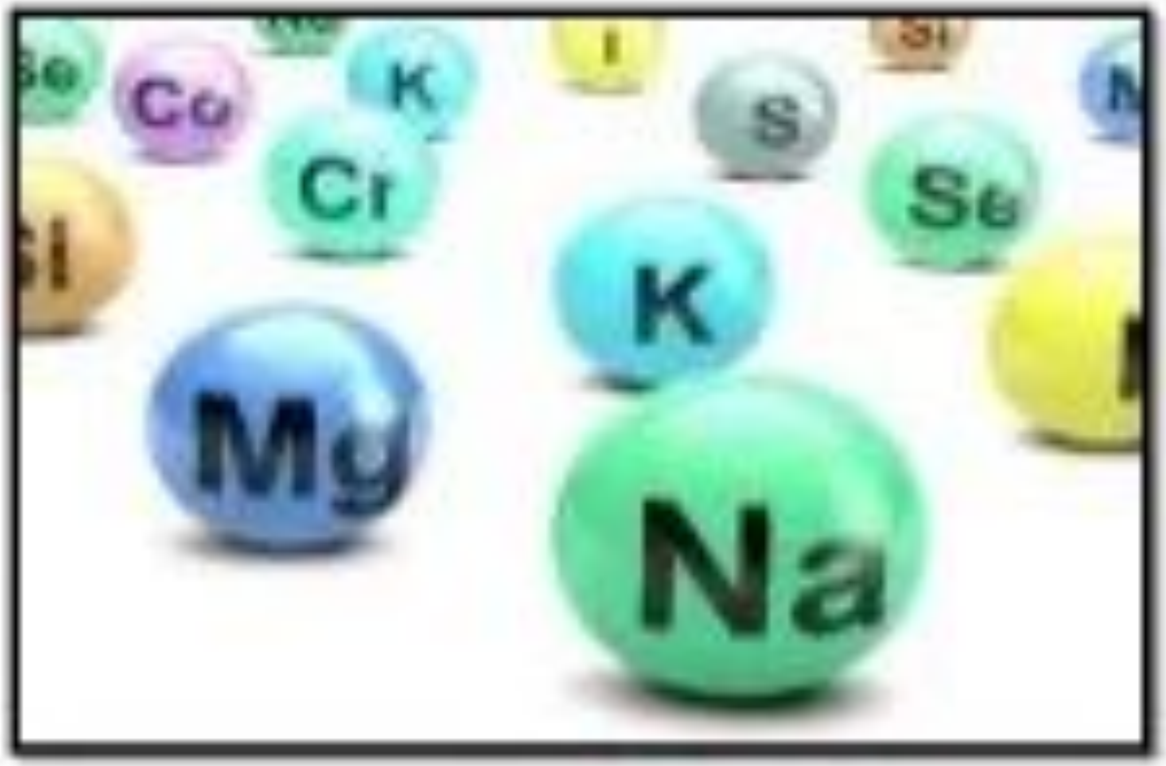
5. Co-enzymes derived from thiamine, biotin, pantothenic acid and lipoic acid also contain sulfur.

6. Sulfates are also important in detoxification mechanism. Phenol, skatole, indole and steroids may be detoxicated in the liver with sulphate ions.





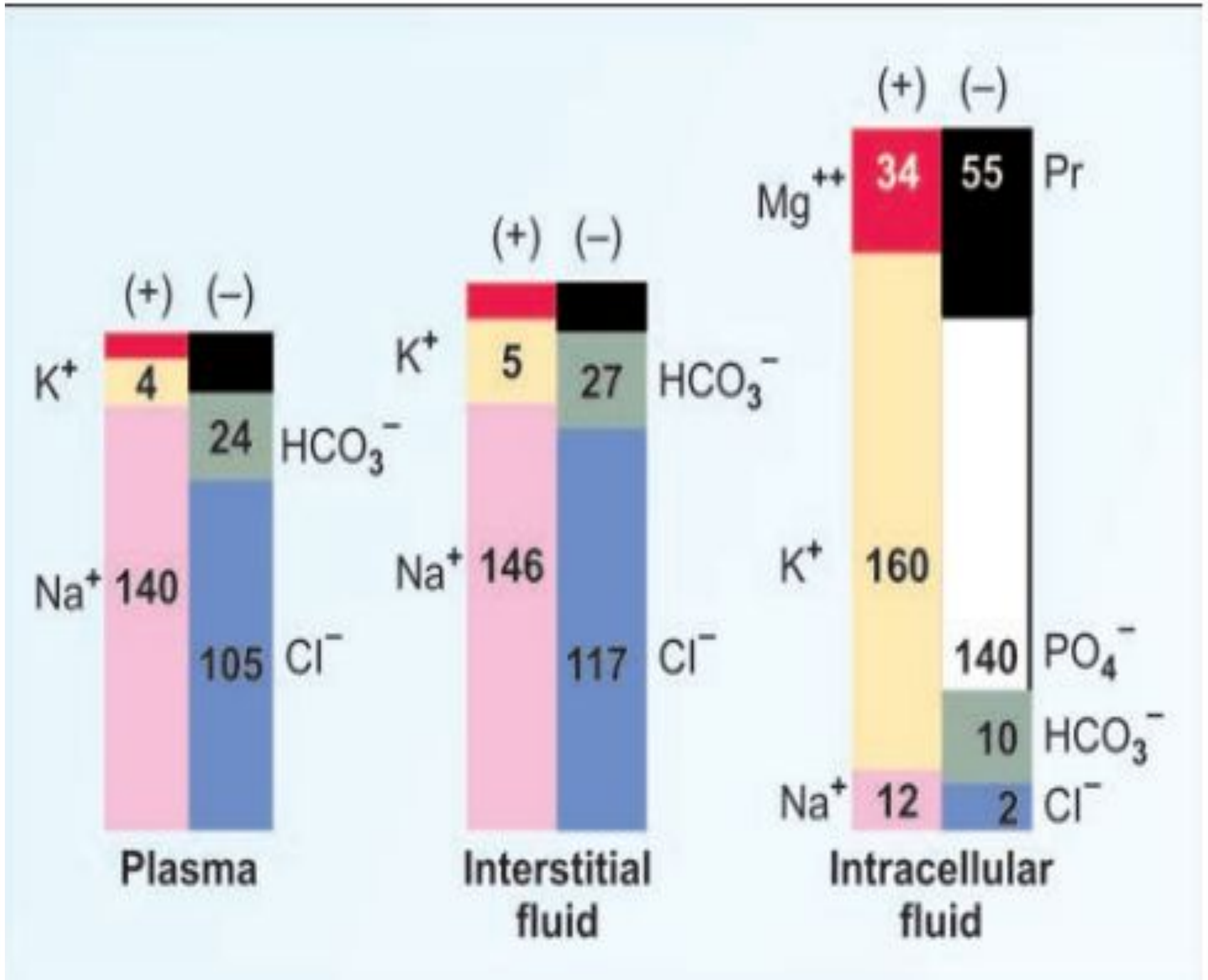
ELECTROLYTES
SODIUM, POTASSIUM & CHLORIDE



Electrolyte concentration of body fluid compartments

Solutes	Plasma mEq/L	Interstitial fluid (mEq/L)	Intracellular fluid (mEq/L)
Cations:			
Sodium	140	146	12
Potassium	4	5	160
Calcium	5	3	–
Magnesium	1.5	1	34
Anions:			
Chloride	105	117	2
Bicarbonate	24	27	10
Sulphate	1	1	–
Phosphate	2	2	140
Protein	15	7	54
Other anions	13	1	–

Gamblegrams showing composition of fluid compartments



SODIUM (Na^+)

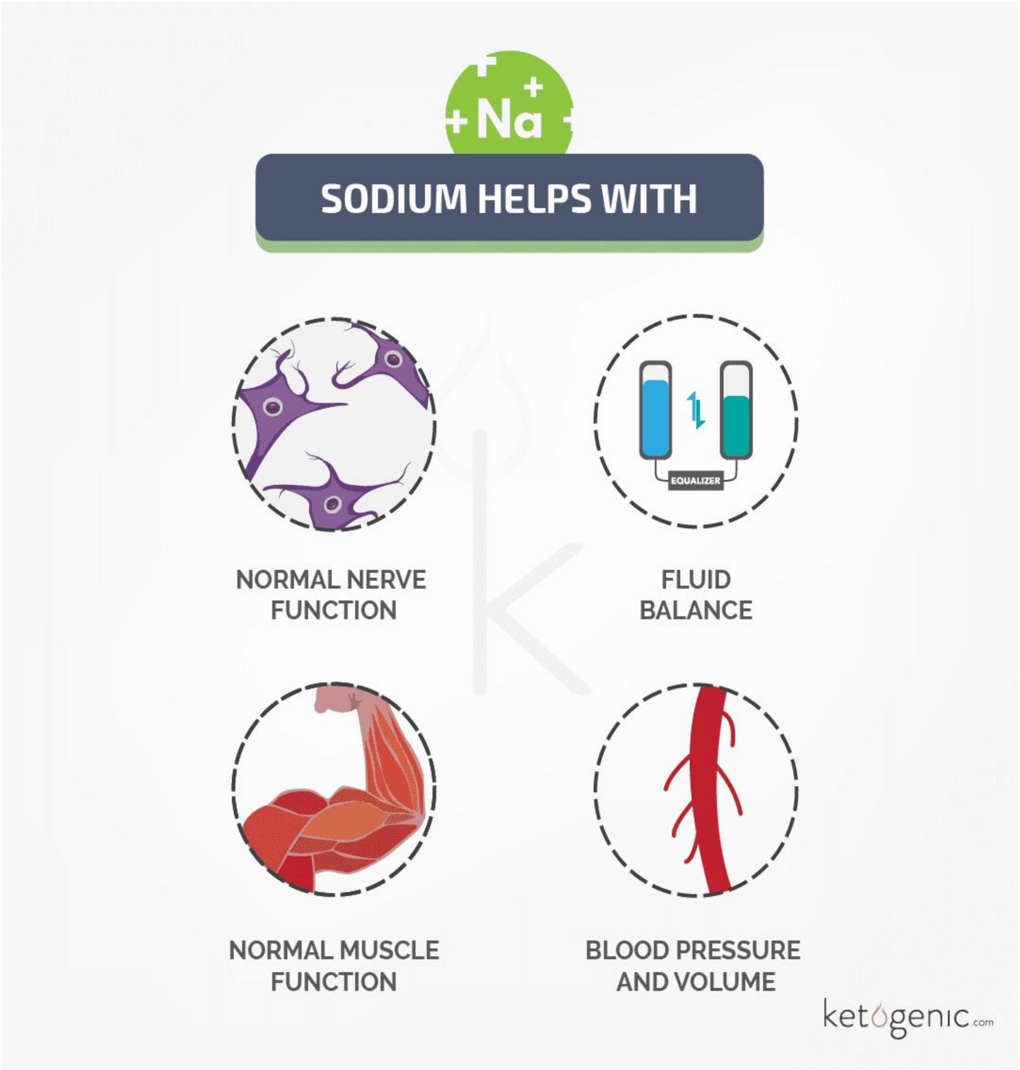
- Sodium is the chief electrolyte which is found in large conc. in extracellular fluid compartment and is one of the most abundant element on Earth.

- The sodium is found in the body mainly associated with chloride as NaCl and $NaHCO_3$.

- ***Sources:*** Sodium is widely distributed in food material. However, major source is **table-salt** used in cooking or seasoning. It is also found in **cheese, butter, processed food, salted nuts, sauces and olives.**

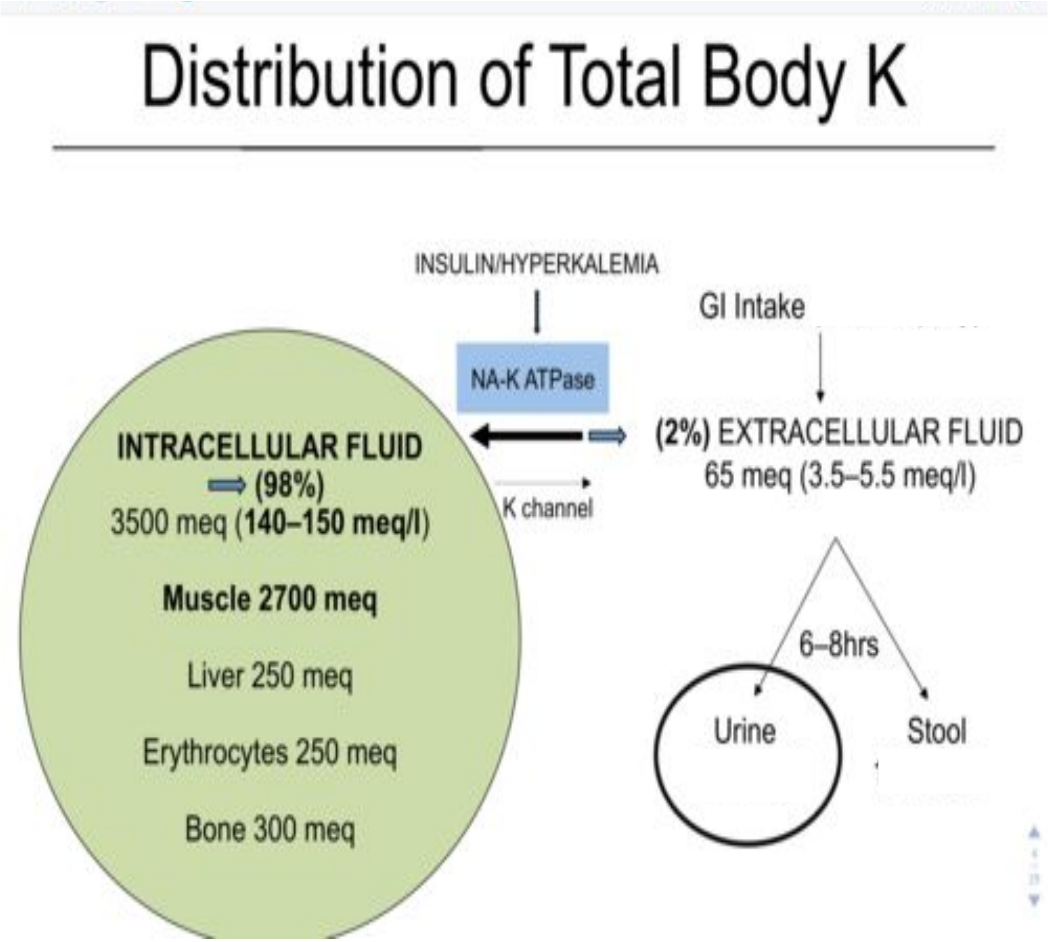
FUNCTIONS

- Maintenance of osmotic pressure and fluid balance: Sodium maintains osmotic pressure of extracellular fluids and helps in retaining water in ECF.
- Neuromuscular excitability.
- Regulates acid-base balance in association with chloride and bicarbonates.
- Role in Action Potential: A local depolarization of nerve or muscle fiber is observed in stimulation. This rapidly increases its permeability to Na+ causing considerable transmembrane influx of Na+.



POTASSIUM (K+)

- Potassium is the major intracellular cation.
- Over 98% is in the intracellular compartment (primarily in the muscle, subcutaneous tissue, and red blood cells) and 2% is in the extracellular compartment.
- The intracellular concentration gradient is maintained by the Na+-K+ ATPase pump.



REQUIREMENT, SOURCES, NORMAL PLASMA LEVEL, EXCRETION and REGULATION

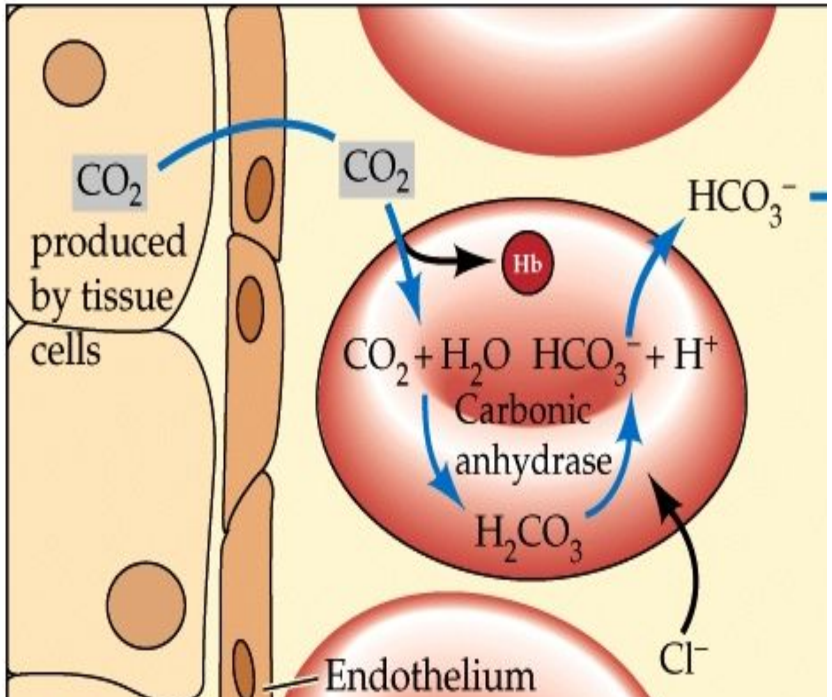
- **Requirement:**
  - Potassium requirement is 3-4 g per day.
- **Sources:**
  - Sources rich in potassium, but low in sodium are **banana, orange, apple, pineapple, almond, dates, beans, yam and potato. Coconut water is a very good source of potassium.**
- **Normal plasma level:**
  - The normal conc. of plasma potassium is 3.5-5 mEq/L.
- **Excretion:**
  - 90% of excess potassium is excreted through kidneys in the urine.
  - The rest 10% potassium is also excreted in gastrointestinal tract, saliva, gastric juice, bile, pancreatic and intestinal juices.

FUNCTIONS

- Many functions of potassium and sodium are carried out in coordination with each other.
  - Nerve conduction
  - Muscular function
  - Osmotic pressure: regulates intracellular osmolality
  - Protein synthesis: K^+ and Mg^{2+} both ions contribute to the stability of various RNA structures
  - Acid-base balance
  - It has an important role in cardiac function

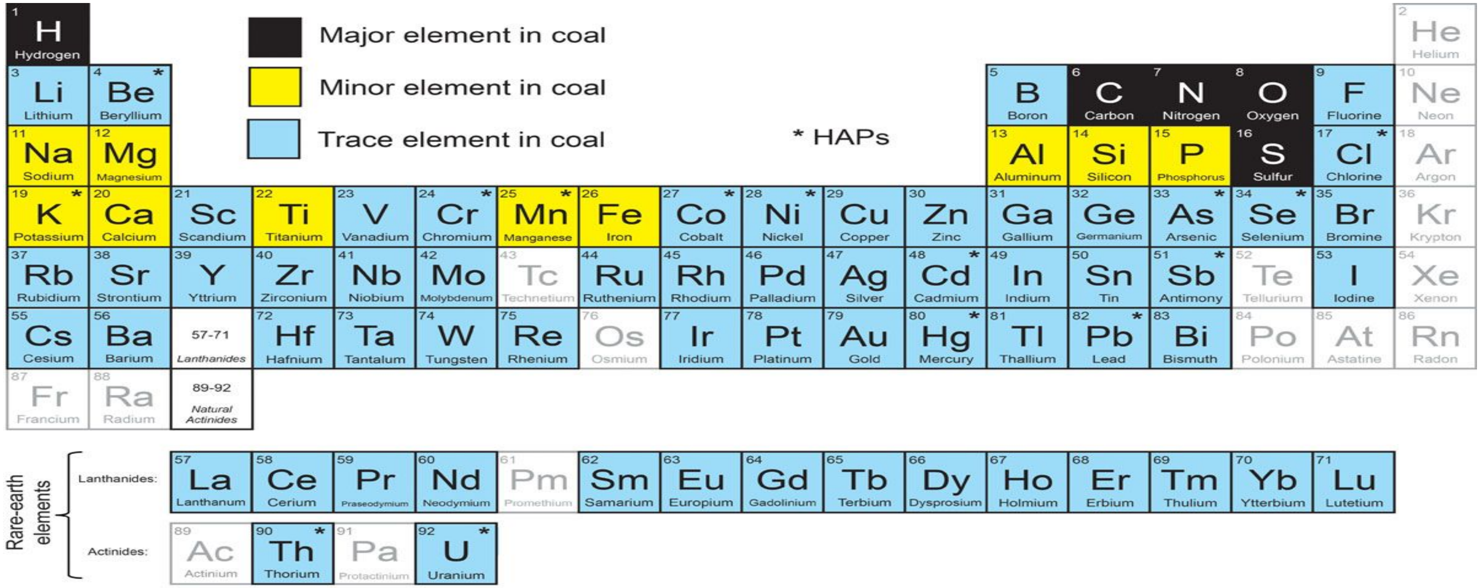
CHLORIDE (Cl–)

- **Functions:**
  - Chloride is important in the formation of hydrochloric acid in gastric juice.
  - Chloride ions are also involved in chloride shift.
- **Sources:**
  - Table salt
  - Vegetables and meats have small proportions of chloride.
  - It is also available in the ‘chlorinated water’ which is normally supplied as a process of purification of water for drinking purpose.
- **Daily Requirement:** About 100-200 mmol is taken in diet as sodium chloride (table salt)



CHLORIDE SHIFT

TRACE ELEMENTS



COPPER (Cu)

- Total body copper is about 100 mg.

- It is seen in muscles, liver, bone marrow, blood, brain, kidney, heart and in hair.

- It occurs as:

1. Erythrocuprein (in red blood cells as colorless)
2. Hepatocuprein (in liver)
3. Cerebrocuprein (in brain).

Copper containing enzymes

- **Ceruloplasmin**
- **Cytochrome oxidase**
- **Cytochrome c**
- **ALA synthase**
- **Monoamine oxidase**
- **Superoxide dismutase**
- **Lysyl oxidase**

ZINC (Zn)

- Total zinc content of body is about 2 gm.

- Distribution:
  - High: In skin, and prostate
  - Average: In bones and teeth
  - Low: In kidneys, muscles, heart, pancreas and spleen and
  - Very low : In brain and lungs

Role in Enzyme Action:

- Zinc forms an integral part of several enzymes (metallo-enzymes) in the body.

- Important zinc containing enzymes are:
    - Carbonic anhydrase, alkaline phosphatase, lactate dehydrogenase, alcohol dehydrogenase and glutamate dehydrogenase.
    - RNA polymerase contains zinc and so it is required for protein biosynthesis.
    - Superoxide dismutase is zinc dependent and so, zinc has antioxidant activity.

OTHER FUNCTIONS

- Insulin when stored in the beta cells of pancreas contains zinc, which stabilizes the hormone molecule thus participate in storage and secretion of Insulin.
- Zinc containing protein in saliva is important for taste sensation.
- Role in Vitamin A Metabolism: Zn^{++} has been claimed to stimulate the release of vitamin A from liver into the blood.
- Zinc is necessary for wound healing. Zinc deficiency delays wound healing.
- Role in Growth and Reproduction: zinc deficiency may lead to ‘dwarfism’ and ‘hypogonadism’. Zinc deficiency also lowers spermatogenesis in males and menstrual cycles are disturbed in females

FLUORIDE (F^-)

- Fluoride is the negative ion of the element fluorine.
- It's widely found in nature, in trace amounts.
- It occurs naturally in air, soil, plants, rocks, fresh water, seawater, and many foods.

Sources:

- Fluoride is derived in human from drinking water.
- Other sources: Tea and salmon contain small amounts of fluoride.

FUNCTIONS

- Fluoride is present in trace amounts in all mineralized tissues of the body such as enamel, dentin, and bone and plays an important role in the mineralization of bones and teeth, helps in tooth development, normal maintenance and hardening of dental enamel and prevention of "dental caries".

- In mineralized tissues and biomaterials, fluoride ions increase the stability of mineralized tissues and materials by decreasing the solubility of hydroxy-apatite mineral.

- The protective effects of fluoride on dental health were first observed in 1930 as there was less tooth decay in communities consuming naturally fluoridated water compared to non-fluoridated areas.

- Due to these beneficial effects of fluoride, it was introduced into dentistry in 1940 and since then, it is being added to various consumer products.

- Greater than 1.2 PPM of fluorine in drinking water of infants/children may increase fluoride contents of the enamel and dentine, may reduce Ca deposition in those tissues and may cause mottling of enamel, in newly erupted permanent teeth: Discoloration, corrosion and stratification of enamel.
- It also promotes normal bone development, increases retention of Ca^{++} and PO_4 and prevent old age osteoporosis.
- High fluoride intakes may raise the fluoride content of bone, stimulate osteoblast activity and cause an abnormal rise in calcium deposition and increased density of bone.

IODINE

- Daily requirement of iodine is 150-200 micrograms/day.
- Its sources are drinking water, fish, cereals, vegetables and iodinated salt.
- Total body contains 25-30 mg of iodine.
- All cells do contain iodine; but 80% of the total is stored in the thyroid gland.
- Upper regions of mountains generally contain less iodine. Such areas are called goiterous belts.

cont..

- The program of iodination of common salt has resulted in increased availability of iodine.

- Ingredients in foodstuffs, which prevent utilisation of iodine are called goitrogens. Goitrogens are seen in maize, millet, bamboo shoots, sweet potatoes and beans. Cabbage contain thiocyanate, which inhibits iodine uptake by thyroid.

- The only biological role of iodine is in formation of thyroid hormones, thyroxine (T4) and tri-iodo thyronine (T3).

MANGANESE

- Manganese is also an essential trace element and is required by the body.
- Acts either as a ‘cofactor’ or as an activator of many enzymes like arginase, isocitrate dehydrogenase (ICD), cholinesterase, lipoprotein lipase, enolase, leucine amino peptidase in intestine, phosphotransferases and 5-oxo-prolinase of kidneys, and small intestine and many others.
- Mn^{++} plays a part in the synthesis or deposition of Mucopolysaccharides (MPS) in the cartilaginous matrices of long bones.

- Mn-deficiency causes significant lowering in the content of chondroitin SO_4.
- Manganese also helps in porphyrin synthesis by participating in δ-ALA synthetase activity.
- Hb-synthesis appears to be depressed in Mn-deficient rats.
- Manganese has been reported also to exhibit "lipotropic effect"(removal of fat from liver) and it stimulates FA synthesis and cholesterol synthesis.
- Manganese also participates in glycoprotein and proteoglycan synthesis

SELENIUM (Se)

- It is widely distributed in all the tissues, highest concentrations are found in liver, kidneys and fingernails. Muscles, bones, blood and adipose tissues show a low concentration of selenium.
- Requirement is 50-100 microgram/day. Normal serum level is 50-100 microgram/dl.
- Glutathione peroxidase (GP) is the important selenium containing enzyme.
- RBC contains good quantity of glutathione peroxidase.
- Thyroxin is converted to T3 by 5'-de-iodinase, which is a selenium containing enzyme.

- In Se deficiency, this enzyme becomes less active, leading to hypothyroidism.
- Selenium concentration in testis is the highest in adult tissue. It is necessary for normal development of spermatozoa.
- It is supplementary to vitamin E and acts as primary antioxidant by scavenging reactive oxygen species and free radical intermediates of polyunsaturated lipid peroxidation.

COBALT

- Cobalt forms an integral part of vitamin B12 and is required as a constituent of this vitamin.
- Cobalt is required to maintain normal bone marrow function and required for development and maturation of red blood cells.
- A deficiency of cobalt results in decreased B12 supply which produces nutritional macrocytic anaemia.
- Cobalt may act as a cofactor for enzyme like glycyl-glycine dipeptidase of intestinal juice.

NICKEL

- Nickel activates several enzyme systems, viz. arginase, carboxylase, trypsin and acetyl-CoA synthetase.

- Nickel is required in trace amount for growth and reproduction.

THANK YOU